Experiment 12

12.1.1 Phone Number Database Management

Algorithm :

Step 1: Start

Step 2: Read n (number of commands)

Step 3: Initialize ph as an empty dictionary {}

Step 4: Set i = 0

Step 5: If i < n, go to Step 6
    Else go to Step 11

Step 6: Read command (cmd)

Step 7: If cmd[0] == "ADD"
        Store ph[cmd[1]] = cmd[2]
    Go to Step 10

Step 8: Else if cmd[0] == "REMOVE"
        Remove cmd[1] from ph (if exists)
    Go to Step 10

Step 9: Else if cmd[0] == "DISPLAY"
        If ph is empty:
            Print "No contacts"
        Else:
            For each name in sorted(ph):
                Print name and ph[name]
    Go to Step 10

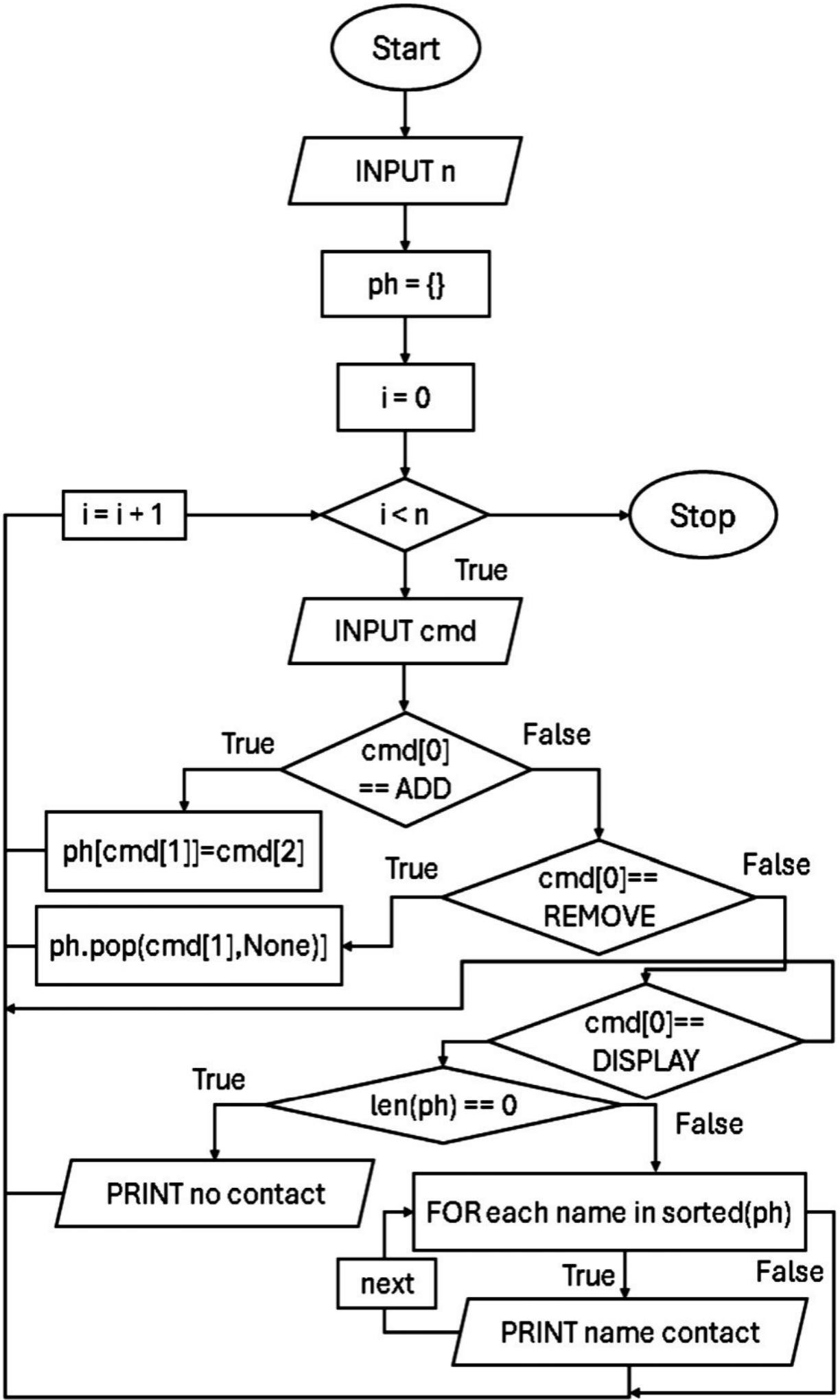
Step 10: Set i = i + 1
Go to Step 5

Step 11: Stop

Code :

```
n=int(input())
ph={}
for i in range(n):
    cmd=input().split()
    if cmd[0] == "ADD":
        ph[cmd[1]] = cmd[2]
    elif cmd[0] == "REMOVE":
        ph.pop(cmd[1], None)
    elif cmd[0] == "DISPLAY":
        if len(ph)==0:
            print("No contacts")
        else:
            for name in sorted(ph):
                print(f"{name}: {ph[name]}")
```

Flowchart :


Start
INPUT n
ph = {}
i = 0
i = i + 1
i < n
Stop
True
INPUT cmd
True
cmd[0]
== ADD
False
ph[cmd[1]]=cmd[2]
True
cmd[0]==
REMOVE
False
ph.pop(cmd[1],None)]
cmd[0]==
DISPLAY
True
len(ph) == 0
False
PRINT no contact
FOR each name in sorted(ph)
next
True
False
PRINT name contact

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12.1.1. Phone Number Database Management

03:16

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
   - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
   - "REMOVE <name>" - name is a string.
   - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:

```
<name>: <phone_number>
```

- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

Sample Test Cases +

Explorer

phoneDat...

SUBMIT

Debugger

```python
# Type Content here...
n=int(input())
ph={}
for i in range(n):
    CMD = input().split()
    if CMD[0]=="ADD":
        ph[CMD[1]]=CMD[2]
    elif CMD[0]=="REMOVE":
        ph.pop(CMD[1],None)
    elif CMD[0]=="DISPLAY":
        if len(ph)==0:
            print("No contacts")
        else:
            for name in sorted(ph):
                print(f"{name}: {ph[name]}")
```

Average time	Maximum time
0.018 s	0.037 s
18.50 ms	37.00 ms

5 out of 5 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 17 ms DEBUG

Expected output

```
4
ADD Alice 9876543210
ADD Bob 8765432109
ADD Charlie 7777777777
DISPLAY
Alice: 9876543210
```

Terminal Test cases

< PREV RESET SUBMIT NEXT >